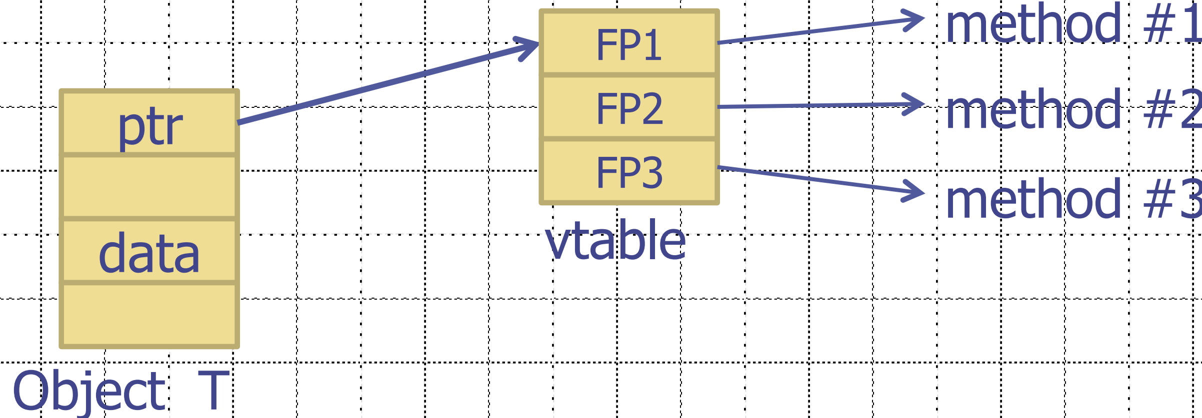
Heap Spray Attacks

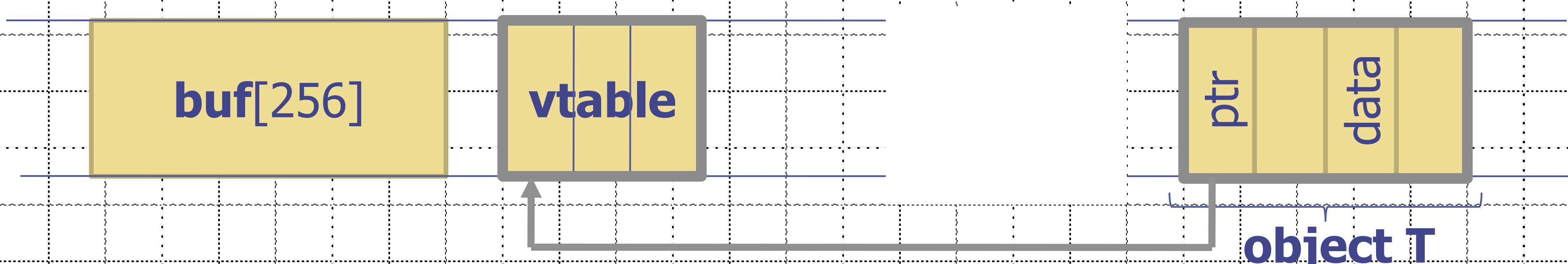
A reliable method for exploiting heap overflows

Heap-based control hijacking

- Compiler generated function pointers (e.g. C++ code)



- Suppose vtable is on the heap next to a string object:



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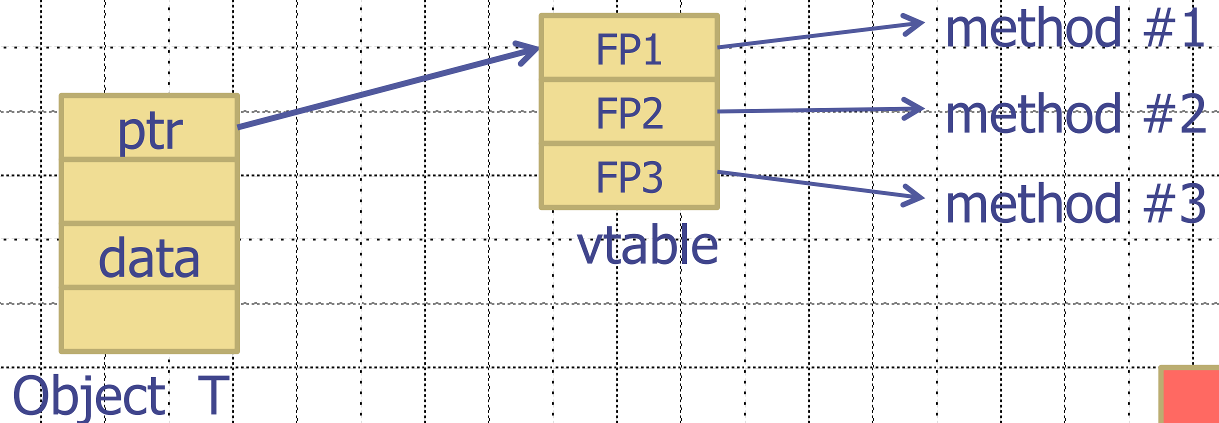
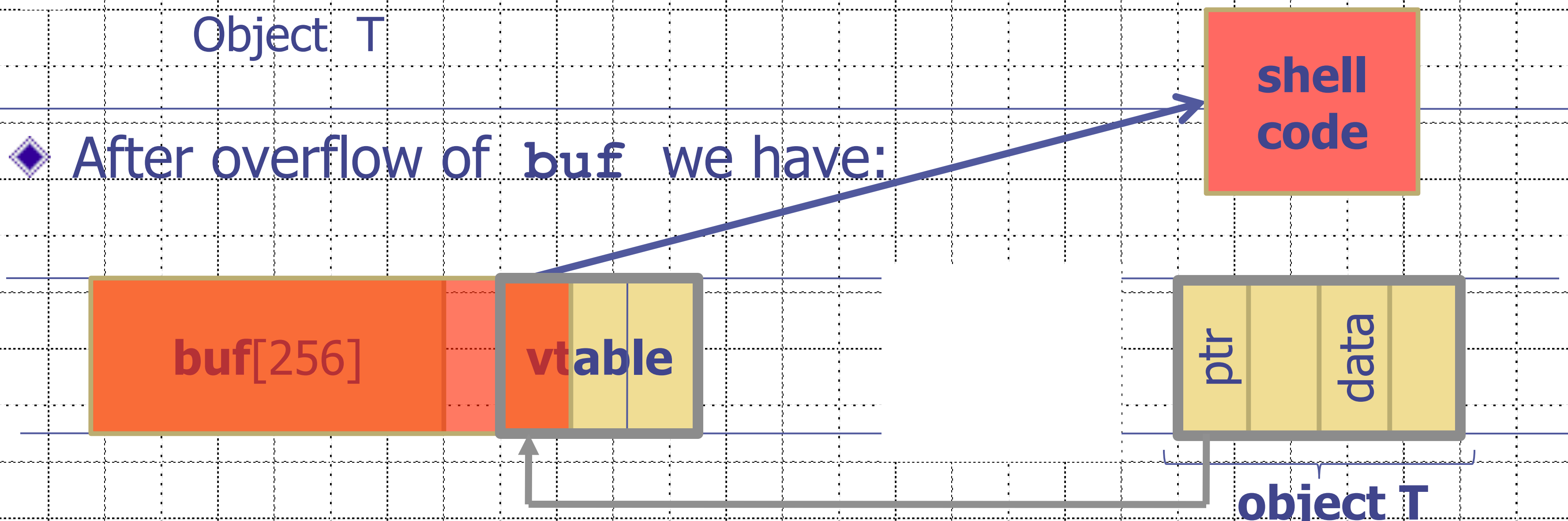


- After overflow of `buf` we have:

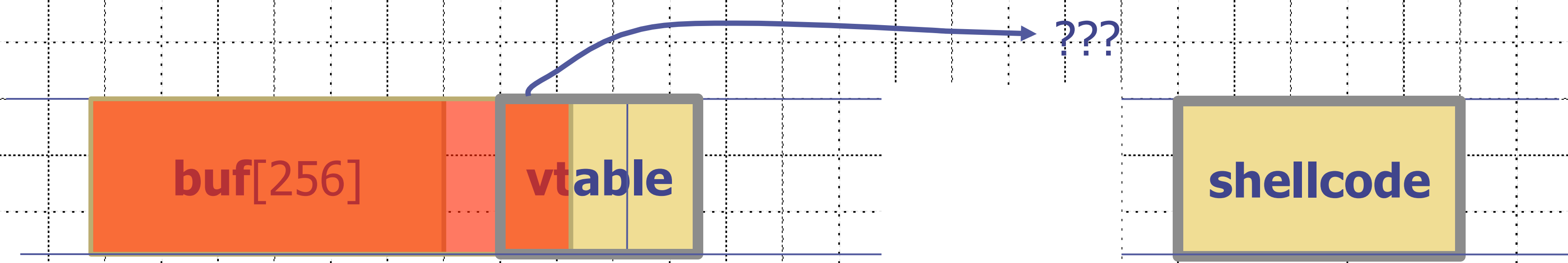


A reliable exploit?

```
<SCRIPT language="text/javascript">
 shellcode = unescape("%u4343%u4343%...");
overflow-string = unescape("%u2332%u4276%...");

cause-overflow( overflow-string );        // overflow  buf[ ]
</SCRIPT>
```

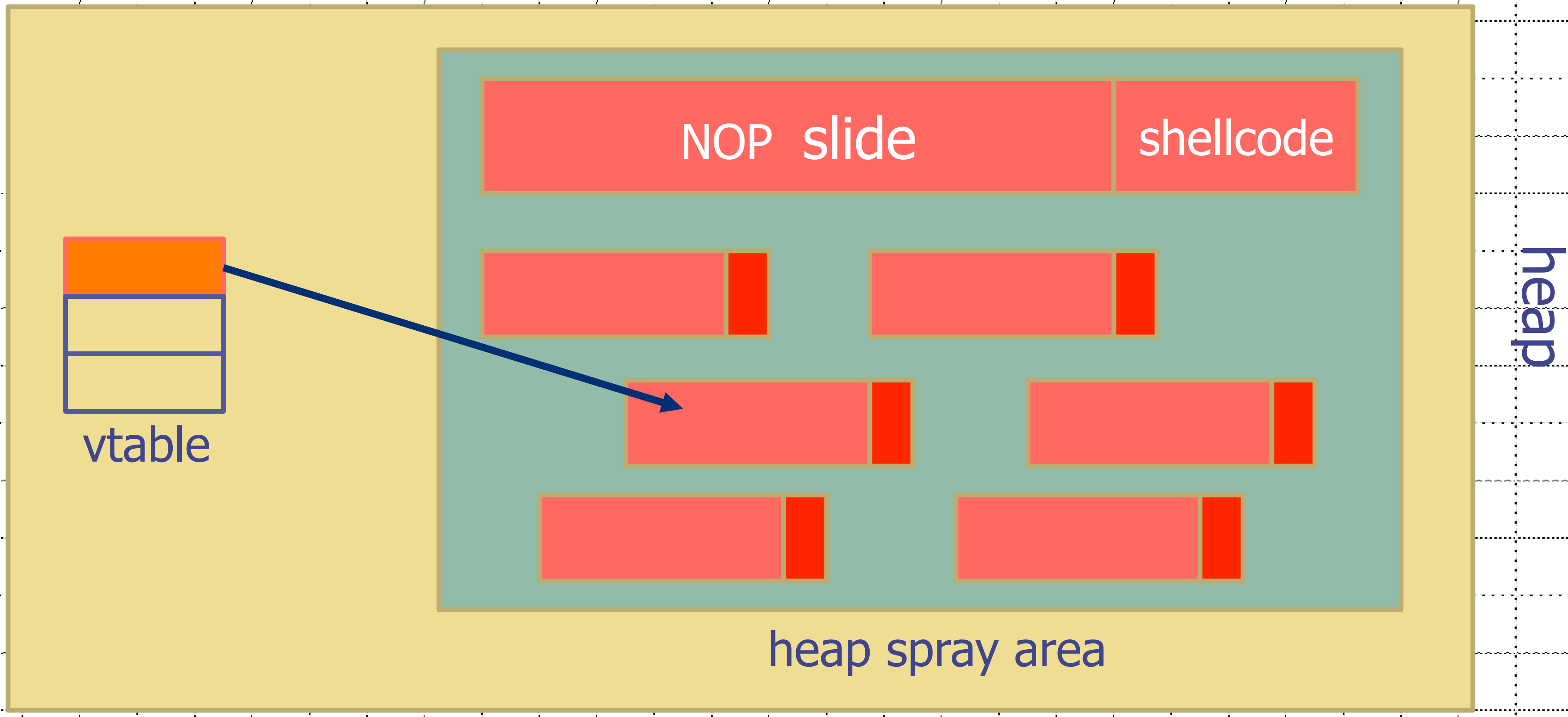
Problem: attacker does not know where browser places **shellcode** on the heap



Heap Spraying [SkyLined 2004]

Idea: 1. use Javascript to spray heap with shellcode (and NOP slides)

2. then point vtable ptr anywhere in spray area



Javascript heap spraying

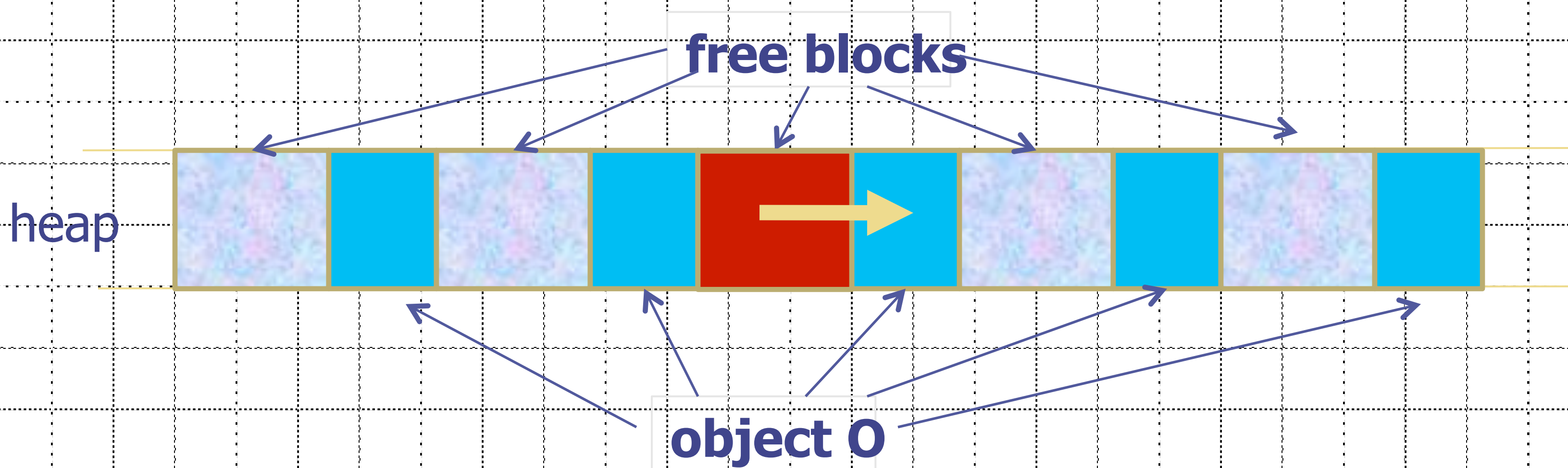
```
var  nop = unescape("%u9090%u9090")
while (nop.length < 0x100000)  nop += nop
var shellcode = unescape("%u4343%u4343%...");
```

```
var x = new Array ()
for (i=0;  i<1000;  i++) {
      x[i] = nop + shellcode;
}
```

- Pointing func-ptr almost anywhere in heap will cause shellcode to execute.

Vulnerable buffer placement

- Placing vulnerable `buf[256]` next to object O:
  - By sequence of Javascript allocations and frees make heap look as follows:



  - Allocate vuln. buffer in Javascript and cause overflow
  - Successfully used against a Safari PCRE overflow [DHM'08]

Many heap spray exploits

[RLZ'08]

Date	Browser	Description
11/2004	IE	IFRAME Tag BO
04/2005	IE	DHTML Objects Corruption
01/2005	IE	.ANI Remote Stack BO
07/2005	IE	javaprxy.dll COM Object
03/2006	IE	createTextRang RE
09/2006	IE	VML Remote BO
03/2007	IE	ADODB Double Free
09/2006	IE	WebViewFolderIcon setSlice
09/2005	FF	0xAD Remote Heap BO
12/2005	FF	compareTo() RE
07/2006	FF	Navigator Object RE
07/2008	Safari	Quicktime Content-Type BO

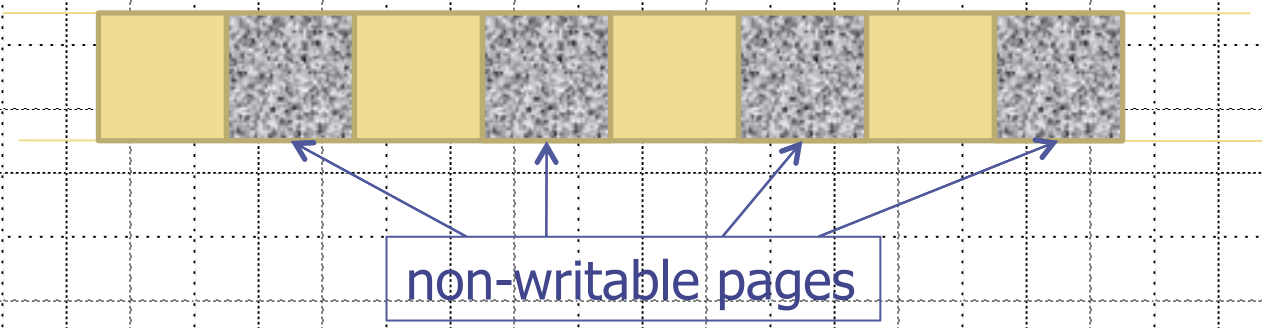
- Improvements: Heap Feng Shui [S'07]
  - Reliable heap exploits **on IE** without spraying
  - Gives attacker full control of IE heap from Javascript

(partial) Defenses

- Protect heap function pointers (e.g. PointGuard)

- Better browser architecture:
  - Store JavaScript strings in a separate heap from browser heap

- OpenBSD heap overflow protection:



prevents cross-page overflows

- Nozzle [RLZ'08] : detect sprays by prevalence of code on heap

References on heap spraying

[1] **Heap Feng Shui in Javascript**,
by A. Sotirov, *Blackhat Europe* 2007

[2] **Engineering Heap Overflow Exploits with JavaScript**
M. Daniel, J. Honoroff, and C. Miller, *WooT* 2008

[3] **Nozzle: A Defense Against Heap-spraying Code Injection Attacks,**
by P. Ratanaworabhan, B. Livshits, and B. Zorn

[4] **Interpreter Exploitation: Pointer inference and JiT spraying**, by Dion Blazakis